Infinite ultrathin annuli refute the universal complementary-error-function plunge profile

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Lane 10 — candidate counterexample for expert review

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Status. Candidate full counterexample to Conjecture 2.6 of Halvdansson. There is a compact regular-closed radial set with finite boundary length for which the asserted uniform $O(R^{-1})$ eigenvalue profile fails. The proof is complete subject to verification of the source's standard uniform-in-k interpretation of the displayed profile.

1 Source conjecture

For the standard Gaussian window $g_0(t) = 2^{1/4}e^{-\pi t^2}$, let $A_\Omega^{g_0}$ be the time-frequency localization operator with symbol $\Omega \subset \mathbb{R}^2$, and write λ_k^Ω for its decreasingly ordered eigenvalues. Conjecture 2.6 of Halvdansson [1] states that, for every compact, regular-closed Ω of finite boundary length,

$$\sup_k \left| \lambda_k^{R\Omega} - \frac{1}{2}\,\mathrm{erfc}\left(\sqrt{2\pi}\frac{k - R^2|\Omega|}{R|\partial\Omega|} \right) \right| = O(R^{-1}). \tag{1}$$

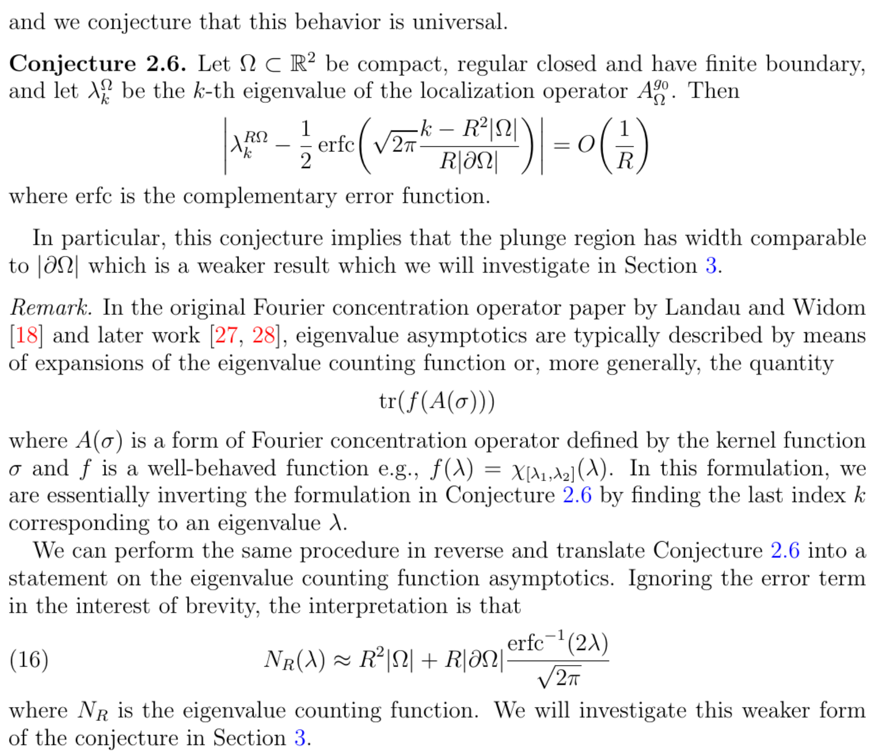
The supremum records the uniform-in-k reading used throughout the source's plunge-profile results and numerical L^∞ comparisons. Without this uniformity, the formula does not describe the moving plunge region $k - R^2|\Omega| = O(R)$.

2 Proof intuition

For a radial symbol, all localization operators are diagonal in the Hermite basis. After the change of variables $s = \pi|z|^2$, the eigenvalue attached to the kth Hermite function is exactly the integral of the Gamma density $e^{-s}s^k/k!$ over the radial symbol. Thus a radial annulus is a short interval sampled by a probability density of width comparable to the square root of its location.

We use infinitely many concentric annuli. Their radii decrease geometrically, so the sum of the lengths of all boundary circles is finite, but their radial widths decrease inductively much faster. At a selected scale R_m, the mth annulus has scaled boundary length tending to infinity, whereas its entire Gamma-density bump has height $o(1)$. Later annuli have negligible total symbol area at that scale. Earlier annuli form a finite radial union, so the source's proved finite-component profile applies to them.

At any fixed threshold $0 < \theta < 1/2$, the full operator therefore has almost the same number of eigenvalues above θ as the finite prefix. The conjectured formula instead charges the invisible tail for all of its boundary circles, creating an unbounded discrepancy. A uniform $O(R^{-1})$ eigenvalue error could create only an $O(1)$ counting discrepancy, which is the contradiction.



and we conjecture that this behavior is universal.

Conjecture 2.6. Let $\Omega \subset \mathbb{R}^2$ be compact, regular closed and have finite boundary, and let λ_k^Ω be the k-th eigenvalue of the localization operator $A_\Omega^{g_0}$. Then

$$\left|\lambda_k^{R\Omega} - \frac{1}{2}\operatorname{erfc}\left(\sqrt{2\pi}\frac{k - R^2|\Omega|}{R|\partial\Omega|}\right)\right| = O\left(\frac{1}{R}\right)$$

where erfc is the complementary error function.

In particular, this conjecture implies that the plunge region has width comparable to $|\partial\Omega|$ which is a weaker result which we will investigate in Section 3.

Remark. In the original Fourier concentration operator paper by Landau and Widom [18] and later work [27, 28], eigenvalue asymptotics are typically described by means of expansions of the eigenvalue counting function or, more generally, the quantity

$$\operatorname{tr}(f(A(\sigma)))$$

where $A(\sigma)$ is a form of Fourier concentration operator defined by the kernel function σ and f is a well-behaved function e.g., $f(\lambda) = \chi_{[\lambda_1,\lambda_2]}(\lambda)$. In this formulation, we are essentially inverting the formulation in Conjecture 2.6 by finding the last index k corresponding to an eigenvalue λ.

We can perform the same procedure in reverse and translate Conjecture 2.6 into a statement on the eigenvalue counting function asymptotics. Ignoring the error term in the interest of brevity, the interpretation is that

$$N_R(\lambda) \approx R^2|\Omega| + R|\partial\Omega|\frac{\operatorname{erfc}^{-1}(2\lambda)}{\sqrt{2\pi}} \tag{16}$$

where N_R is the eigenvalue counting function. We will investigate this weaker form of the conjecture in Section 3.

Figure 1: Conjecture 2.6 and its eigenvalue-counting interpretation on printed page 13 of the source paper.

3 Radial diagonalization and two lemmas

Put

$$p_k(s) := e^{-s}\frac{s^k}{k!} \qquad (k = 0, 1, \ldots).$$

If $E \subset [0, \infty)$ and

$$\Omega_E := \{z \in \mathbb{R}^2 : \pi|z|^2 \in E\},$$

then the Hermite functions diagonalize $A^{g_0}_{R\Omega_E}$, and its unordered eigenvalues are

$$\mu_k(R, E) = \int_{R^2 E} p_k(s)\, ds. \tag{2}$$

Indeed, $|V_{g_0} h_k(z)|^2 = e^{-\pi|z|^2}(\pi|z|^2)^k/k!$, and $s = \pi|z|^2$ transforms planar radial measure into ds. This is the same radial formula used in the disk and annulus calculations of the source [1, Section 2].

Lemma 1 (A thin annulus has a small spectral bump)**.** *There is an absolute constant C_0 such that, whenever $a > 0$, $\ell > 0$, $R^2 a \geq 1$, and $I = [a, a + \ell]$,*

$$\sup_{k \geq 0} \int_{R^2 I} p_k(s)\, ds \leq C_0 \frac{R\ell}{\sqrt{a}}.$$

Proof. For fixed s, $p_k(s)$ is the probability that a Poisson random variable of mean s equals k. Unimodality in k and Stirling's lower bound for $\lfloor s \rfloor!$ give the standard maximal-atom estimate

$$\sup_{k \geq 0} p_k(s) \leq C_0 s^{-1/2} \qquad (s \geq 1).$$

The interval $R^2 I$ has length $R^2\ell$ and lies to the right of $R^2 a$. Integrating the displayed estimate gives $C_0 R^2\ell/(R\sqrt{a})$. □

For $A, P > 0$, define the model profile

$$F_{R,A,P}(x) := \frac{1}{2}\operatorname{erfc}\left(\sqrt{2\pi}\frac{x - R^2 A}{RP}\right), \qquad b_\theta := \frac{\operatorname{erfc}^{-1}(2\theta)}{\sqrt{2\pi}}.$$

Thus $b_\theta > 0$ for $0 < \theta < 1/2$.

Lemma 2 (Uniform profile implies bounded counting error)**.** *Let $(\lambda_k(R))$ be decreasing. If*

$$\sup_k |\lambda_k(R) - F_{R,A,P}(k)| \le C/R,$$

then, for each fixed $\theta \in (0,1)$,

$$\#\{k : \lambda_k(R) > \theta\} = R^2 A + RPb_\theta + O(1). \tag{3}$$

The $O(1)$ constant may depend on $C, P,$ and θ, but not on R.

Proof. The count is sandwiched between the numbers of integers k for which $F_{R,A,P}(k) > \theta + C/R$ and $F_{R,A,P}(k) > \theta - C/R$. Solving these two inequalities for k gives

$$k < R^2 A + RP\frac{\operatorname{erfc}^{-1}(2\theta + O(R^{-1}))}{\sqrt{2\pi}}.$$

The inverse complementary error function is continuously differentiable near 2θ. Its perturbation is $O(R^{-1})$, which changes the cutoff by $O(1)$. Integer rounding contributes another $O(1)$. □

4 The infinite-annulus symbol

Set

$$a_m := 4^{-m}, \qquad c_m := a_m \qquad (m \ge 1).$$

We choose positive lengths $\ell_m < a_m/10$ inductively. Put

$$R_m := \frac{c_m\sqrt{a_m}}{\ell_m}. \tag{4}$$

The lengths can be chosen to have all of the following properties:

$$R_m \longrightarrow \infty, \qquad\qquad R_m\sqrt{a_m} \longrightarrow \infty, \tag{5}$$

$$R_i^2 \sum_{j>i} \ell_j \le c_i \qquad\qquad (i \ge 1), \tag{6}$$

and the finite-prefix counting constants introduced below are $o(R_m\sqrt{a_m})$.

Here is the elementary induction. After $\ell_1, \dots, \ell_{m-1}$ have been chosen, the union of the first $m-1$ annuli is fixed. Proposition 2.5 of the source gives its uniform $O(R^{-1})$ profile, and Lemma 2 gives a finite counting constant, uniformly for thresholds in a compact subinterval of $(0,1)$. Choose ℓ_m small enough that the two quantities in (5) exceed any prescribed bounds and that this fixed finite-prefix constant is at most $2^{-m}R_m\sqrt{a_m}$. Also impose

$$\ell_m \le 2^{-m}\min_{i<m}\frac{c_i}{R_i^2}.$$

For each fixed i, summing this last condition over future m yields (6). All requirements are compatible because ℓ_m may be arbitrarily small.

Define

$$E := \{0\} \cup \bigcup_{m \geq 1} [a_m, a_m + \ell_m], \qquad \Omega := \Omega_E. \tag{7}$$

The intervals are disjoint and accumulate only at 0. Hence E and Ω are compact. The interior of Ω is the union of the open annuli, and its closure is exactly Ω; thus Ω is regular closed. Its area and boundary length are

$$|\Omega| = \sum_m \ell_m < \infty, \qquad |\partial\Omega| = 2\sqrt{\pi} \sum_m \big(\sqrt{a_m} + \sqrt{a_m + \ell_m}\big) < \infty. \tag{8}$$

The accumulation point at the origin has zero $\mathcal{H}^1$ measure. In particular, (7) satisfies every geometric hypothesis stated in Conjecture 2.6.

5 Failure of the conjectured counting law

Fix $\theta \in (0, 1/2)$, for example $\theta = 1/4$. Let $E_{<m} := \bigcup_{j<m} [a_j, a_j + \ell_j]$, and write $A_{<m}$ and $P_{<m}$ for the area and boundary length of $\Omega_{E_{<m}}$. This is a finite union of annuli. Proposition 2.5 of the source and Lemma 2 give, at $R = R_m$ and uniformly for $u \in [\theta/2, \theta]$,

$$N_{<m}(u) = R_m^2 A_{<m} + R_m P_{<m} b_u + O_m(1), \tag{9}$$

where the construction makes $O_m(1) = o(R_m \sqrt{a_m})$.

Let $\tau_{m,k}$ be the contribution in (2) of the annuli with indices $j \geq m$. Since $R_m \ell_m / \sqrt{a_m} = c_m$ and $R_m^2 a_m \geq 1$ for large m, Lemma 1 bounds the mth contribution by $C_0 c_m$. For the later annuli we use only $p_k \leq 1$ and (6):

$$\sup_k \sum_{j>m} \int_{R_m^2 [a_j, a_j + \ell_j]} p_k(s)\, ds \leq R_m^2 \sum_{j>m} \ell_j \leq c_m.$$

Consequently

$$0 \leq \tau_{m,k} \leq \delta_m := (C_0 + 1) c_m \qquad \text{for every } k. \tag{10}$$

Because all pieces are diagonal in the same Hermite basis, (10) immediately gives

$$N_{<m}(\theta) \leq N_{\Omega, R_m}(\theta) \leq N_{<m}(\theta - \delta_m) \tag{11}$$

for all large m. Since b_u is Lipschitz near θ, (9) and (11) imply

$$N_{\Omega, R_m}(\theta) \leq R_m^2 A_{<m} + R_m P_{<m} b_\theta + C_1 R_m P_{<m} c_m + o(R_m \sqrt{a_m}). \tag{12}$$

On the other hand, the count forced by the conjectured profile is

$$Q_m := R_m^2 |\Omega| + R_m |\partial\Omega| b_\theta.$$

The mth annulus alone has boundary length

$$2\sqrt{\pi}\big(\sqrt{a_m} + \sqrt{a_m + \ell_m}\big) \geq 4\sqrt{\pi a_m}.$$

Subtracting (12) from Q_m, discarding the nonnegative tail-area term, and using $P_{<m} \le |\partial\Omega|$ gives

$$
\begin{aligned}
Q_m - N_{\Omega,R_m}(\theta) &\ge 4\sqrt{\pi}\, b_\theta R_m \sqrt{a_m} - C_1 |\partial\Omega| R_m c_m - o(R_m\sqrt{a_m}) \\
&= R_m\sqrt{a_m}\left(4\sqrt{\pi}\, b_\theta - C_1|\partial\Omega| \frac{c_m}{\sqrt{a_m}} - o(1)\right).
\end{aligned}
$$

But $c_m/\sqrt{a_m} = \sqrt{a_m} \to 0$, while $R_m\sqrt{a_m} \to \infty$. Therefore

$$
Q_m - N_{\Omega,R_m}(\theta) \longrightarrow +\infty. \tag{13}
$$

Theorem 1. *For the compact regular-closed radial set Ω in (7), whose boundary has finite length,*

$$
\limsup_{R\to\infty} R \sup_k \left| \lambda_k^{R\Omega} - \frac{1}{2}\operatorname{erfc}\left(\sqrt{2\pi}\frac{k - R^2|\Omega|}{R|\partial\Omega|}\right)\right| = +\infty.
$$

In particular, Conjecture 2.6 is false as stated.

Proof. If the displayed limsup were finite, the conjectured uniform $O(R^{-1})$ profile would hold. Lemma 2, applied to the fixed set Ω, would then give $N_{\Omega,R_m}(\theta) = Q_m + O(1)$. This contradicts the unbounded gap (13). □

6 Scope, verification, and novelty bounds

The example exploits infinitely many boundary components at vanishing scales. It does not contradict the source's Proposition 2.5 for finitely many radial components. Nor does it rule out a corrected universality theorem for smooth, uniformly Ahlfors-regular, or finitely-component boundaries. The source itself notes immediately after Conjecture 2.6 that stronger boundary regularity may be necessary.

The phrase "have finite boundary" is read as finite boundary length, consistent with the quantity $|\partial\Omega|$ in the formula. This reading is also distinguished in the source from the explicit "finite number of connected components" hypothesis in Proposition 2.5. If the conjecture was intended to assume finitely many boundary components, the present example is a statement correction rather than a counterexample to that stronger version.

The proof is non-computational. The checks isolated in the argument are:

1. the radial Hermite eigenvalue formula (2);
2. the Poisson maximal-atom estimate in Lemma 1;
3. compactness, regular closedness, and the finite perimeter sum (8);
4. feasibility of the diagonal induction defining ℓ_m;
5. the threshold sandwich (11); and
6. the profile-to-counting implication in Lemma 2.

A bounded novelty search on 9 August 2026 covered the run's registry, solution, attempt, and proof-gap indexes; arXiv:2502.11805 and the exact paper title; searches combining "boundary universality," "localization operator," "erfc," and the displayed scaling; and the author's 2025 thesis. The thesis still presents the formula as a conjecture. No later explicit resolution or infinite-annulus counterexample was found. This is not an exhaustive MathSciNet or zbMATH priority search.

Human-review recommendation. High priority. Verify the uniformity in the source's finite-annulus Proposition 2.5, the inductive domination of its prefix constants, and the convention that finite boundary means finite $\mathcal{H}^1$ length. The remaining estimates are elementary and exact.

References

[1] S. Halvdansson, *Empirical plunge profiles of time-frequency localization operators*, arXiv:2502.11805 (2025), Conjecture 2.6 and Proposition 2.5. https://arxiv.org/abs/2502.11805.